PROJECT-BASED LEARNING

PHASE-II EVALUATION

Home Energy Analytics and Budget Advisor

Team ID: DSCPP-III-2026-T127

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID:	DSCPP-III-2026-T127
Semester:	3rd
Section:	ML4
Domain:	C++ data processing with HTML/CSS/JavaScript GUI

Team Members

1. Ruhani Aneja – Team Lead
2. Suryansh Tayal – Team Member
3. Aditi Gupta – Team Member

Interactions completed: 5

Mentor

Ms. Shrishti Chamoli

Brief Discussion: Five mentor interactions were completed and feedback was positive. Phase-I focused on the project topic and its details; Phase-II adds the GUI and code.

Phase-I Recap & Feedback Incorporated

Phase-I Status

The project was proposed as a household electricity analysis tool. The current project description includes a predefined CSV dataset containing 100 appliance records and calculations for estimated energy use and cost.

Feedback Received

- Mentor feedback was positive.
- Continue with the selected household energy problem.
- Show concrete progress beyond the Phase-I topic proposal.

Changes Incorporated

- Added a working GUI with dashboard views.
- Added code and data-processing work.
- Added appliance, energy, budget and insight screens.

Brief Discussion: Phase-I contained project-topic details. Phase-II progress now includes the GUI and code; the team completed five mentor interactions and received positive feedback.

Problem Statement & Finalized Objectives

Problem Statement

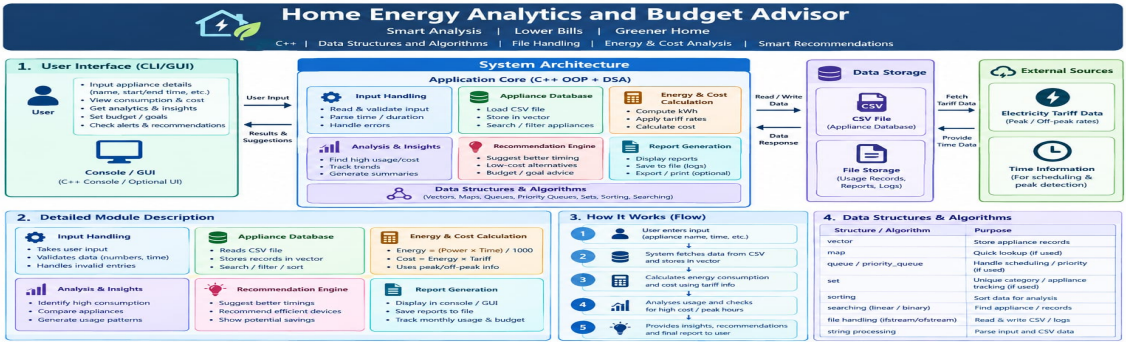
Households use appliances with different power ratings and operating patterns. Comparing the estimated energy contribution of many devices can be difficult. The project provides a consolidated, dataset-driven view of estimated household energy consumption and electricity cost.

The application uses predefined values and assumed operating hours. It does not represent live smart-meter monitoring or physical appliance control.

Finalized Objectives

1. Load appliance records from a predefined CSV dataset.
2. Estimate energy consumption from appliance power and daily operating hours.
3. Estimate electricity cost using a configured tariff per kWh.
4. Present analysis and results supported by the current implementation.

Updated System Architecture / Workflow



Major Components: Input handling, appliance data, energy/cost calculation, analysis, recommendations and report output.

Changes Since Phase-I: The initial topic proposal has progressed to GUI screens and code/data-processing work. The system uses simulated data, not live smart-meter readings.

Updated System Architecture / Workflow – Data Flow Details

Application Workflow

1. Start the application or select an operation, depending on the current program.
2. Load the predefined CSV file.
3. Parse the record fields and validate the values supported by the code.
4. Store appliance records in the program's in-memory structure.
5. Calculate estimated energy and electricity cost.
6. Display the analysis available in the current build.

Data Fields

The dataset is described as containing 100 appliance records with these seven fields:

- `id`
- `name`
- `category`
- `power_watts`
- `hours_per_day`
- `priority`
- `status`

Module Design & Functional Requirements

Major Modules

1. CSV data loading
2. Appliance record handling
3. Energy and cost calculation
4. Analysis and results

Functional Requirements

- Read the predefined CSV file.
- Parse appliance fields.
- Estimate energy from power and usage hours.
- Estimate cost from energy and tariff.

Module Dependency

Energy and cost calculations depend on appliance records being loaded and parsed. Cost estimation also depends on the configured electricity tariff.

Brief Discussion: Explain the purpose of each module and confirm which functions are present in the current application.

Module Design & Functional Requirements – Calculations

Daily Energy Calculation

$$\text{Daily Energy (kWh)} = \frac{\text{Power (W)} \times \text{Hours per day}}{1000}$$

Period Energy Calculation

$$\text{Period Energy (kWh)} = \frac{\text{Power (W)} \times \text{Hours per day} \times \text{Days}}{1000}$$

Estimated Electricity Cost

$$\text{Cost} = \text{Estimated Energy (kWh)} \times \text{Tariff per kWh}$$

Worked Example: AC_1

Power = 1,500 W; daily use = 6 hours; period = 30 days.

$$\frac{1500 \times 6 \times 30}{1000} = 270 \text{ kWh}$$

At an illustrative tariff of Rs. 6/kWh, estimated cost is Rs. 1,620. Replace this tariff with the value configured in the actual code.

Technology Stack & Development Methodology

Technology Stack

- Programming: C++
- Dataset: CSV file
- File input: C++ file streams, where implemented
- Data processing: string parsing, where implemented
- Version Control: Git / GitHub
- GUI: HTML, CSS and JavaScript

Development Methodology

1. Requirement Analysis
2. System Design
3. Module Development
4. Integration
5. Testing & Validation

Brief Discussion: The project combines C++ processing with an HTML/CSS/JavaScript interface. The supplied report describes a local CSV workflow; testing remains partial.

Implementation of PBL Course Concepts

Course	Concepts Actually Implemented	Project Module
Data Structures in C/C++	CSV record processing and analysis; confirm exact containers and algorithms in source	Appliance records / analysis
OOPs with C++	C++ program structure and data processing; list OOP features only when verified	Confirm from source code
Operating Systems	No specific OS concept is confirmed in the report	To verify
DBMS	CSV file is used as the data source; no SQL database is confirmed	CSV data loading

Important: Mention only concepts that have actually been implemented in the project.
Brief Discussion: Explain file handling, string processing and data structures only where they appear in the current source code. No unverified OS or SQL concept is claimed.

Implementation of PBL Course Concepts – Where They Fit

File Handling

The project description identifies a predefined CSV dataset as the source of appliance records. If the current code reads the file using C++ file streams, explain the file-opening, reading and closing operations.

String Processing

If implemented, explain how each CSV line is separated into fields and how the values are converted into the required data types.

Data Structures

Name the exact container used by the program, such as an array, vector or map, only after checking the source code. Explain how the records are stored and traversed.

Algorithms

Describe linear search, sorting, category aggregation or maximum lookup only if the corresponding algorithm exists in the code. Do not claim an algorithm merely because it is possible to add it.

Development & Implementation Progress

Module	Status	Work Completed
Data and processing	In progress / version check	Report describes a 100-record CSV; GUI screenshots display 20 appliances over 30 days.
Energy and cost views	Demonstrated in GUI	Appliance energy, estimated cost, peak-hour and category views are visible in screenshots.
Budget and insight views	Demonstrated in GUI	Budget status, weekly analysis and recommendations are shown.
Testing and validation	Partial	Some testing has been completed; edge cases and manual calculation checks still need recording.

Overall Project Completion: 70%

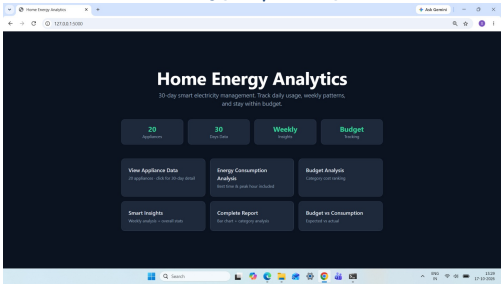
Brief Discussion: Overall progress is estimated at 70%. The GUI screens are available; testing is partial and the dataset/version difference must be reconciled before final submission.

Working Prototype / Demonstration

Implemented Features

- Appliance data view
- Energy and cost analysis
- Budget status and category costs
- Smart insights and report views

Prototype / Output

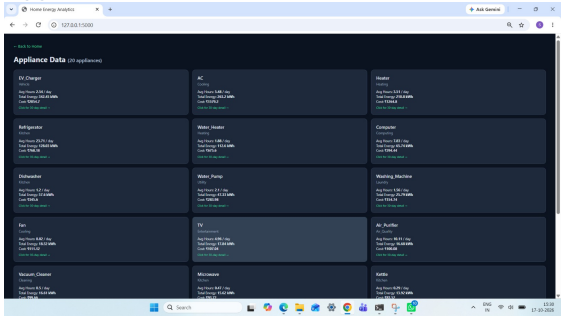


Home dashboard: appliance data, energy analysis, budget, insights and reports.

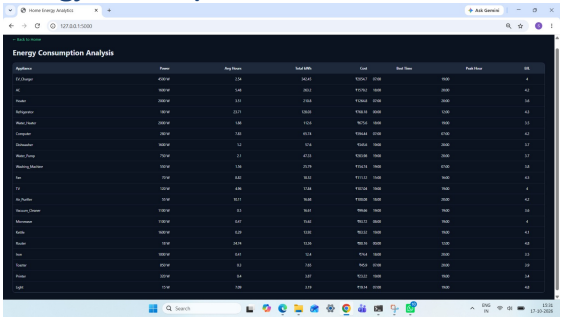
Brief Discussion: Demonstrate the actual working implementation. The screenshot is from the current GUI. C++/CSV integration should be demonstrated during evaluation.

Working Prototype / Demonstration – Sample Output

Appliance Data



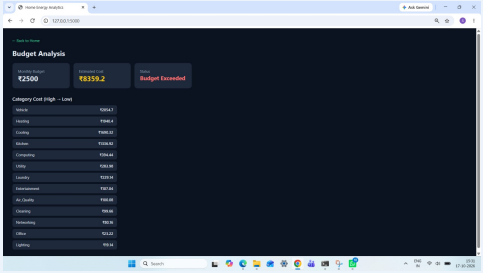
Energy Consumption



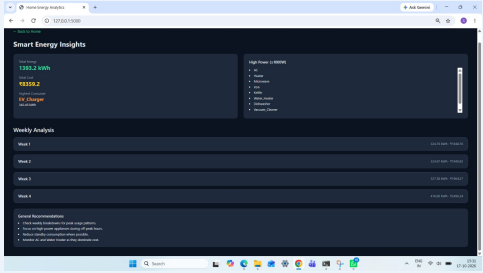
Observed GUI scope: 20 appliances and 30-day analysis. Values shown are the GUI's displayed estimates; they should be checked against the current CSV and calculation code before being presented as validated test results.

Budget Analysis & Smart Energy Insights

Budget Analysis



Smart Energy Insights



Displayed values: Budget screen shows a monthly budget of Rs. 2,500 and estimated cost of Rs. 8,359.2. Insights screen shows 1,393.2 kWh total energy and weekly analysis. These values are transcribed from the supplied GUI screenshots; calculation validation is still partial.

Testing, Results & Validation

Test / Module	Expected Result	Observed Result	Status
GUI data view	Appliance list displayed	Visible in supplied screenshot	Demonstrated
Energy/cost analysis	Values shown for 30 days	Visible in supplied screenshot; manual check pending	Partial
Insights / categories	Summary and categories shown	Visible in supplied screenshot	Demonstrated
Missing / malformed CSV	Safe error handling	Not yet documented	Pending
Budget comparison	Budget status shown	Visible in supplied screenshot; manual check pending	Partial

Results / Validation

GUI screens are available as evidence of the displayed pages. Since testing is partial, missing/malformed CSV handling and manual formula checks remain pending; do not label these as passed.

Testing, Results & Validation – Calculation Check

Manual Calculation

For an appliance rated at 1,500 W, used 6 hours per day for 30 days:

$$E = \frac{1500 \times 6 \times 30}{1000} = 270 \text{ kWh}$$

At an illustrative tariff of Rs. 6/kWh:

$$\textit{Cost} = 270 \times 6 = \text{Rs. } 1620$$

Validation Checklist

- Compare the program output with the manual calculation.
- Confirm that the selected period and usage hours are applied correctly.
- Confirm the tariff value used by the current program.
- Record actual outputs and any errors.
- Capture evidence of each completed test.

The tariff above is illustrative. Use the actual configured value in final results.

GitHub Repository & Development Activity

Repository Information

- GitHub Link: `https://github.com/ruhanianeja29/Electricity-consumption-analyzer`
- README.md: Verify current status
- Contributors: Verify all members
- Repository status: Verify latest upload
- Branches: If applicable

Development Activity

- Repository link is available
- Main source file: `main.cpp`
- Dataset: `electricity_management_data.csv`
- Branch and commit details need checking
- Do not report unverified commit counts

Requirement

All team members must be contributors and commit individually using their own GitHub accounts.

Brief Discussion: Show repository activity, README, contributor list and commit history during the presentation.

Individual Contribution

Member	Role	Technical Contribution	GitHub Commits
Ruhani Aneja	Team Lead	Report writing, coordination and GUI screenshots.	Verify
Suryansh Tayal	Team Member	C++ coding and data processing.	Verify
Aditi Gupta	Team Member	Initial ideas and planning.	Verify

Individual Evaluation

Every member should be prepared to explain and demonstrate their own technical contribution.

Brief Discussion: These roles are based on the team-confirmed division of work. Individual GitHub commit counts have not been verified and should be checked before evaluation.

Individual Contribution – Evidence

Evidence for Each Member

- Name the feature, module or task personally completed.
- Identify the relevant source-code function or file.
- Explain the logic and demonstrate it during evaluation.
- Show the member's own GitHub commits.
- State the exact contribution without claiming another member's work.

Team-confirmed roles

Ruhani Aneja	Report writing, coordination and GUI screenshots
Suryansh Tayal	C++ coding and data processing
Aditi Gupta	Initial ideas and planning

GitHub commit counts and per-member commit history remain to be verified from the repository.

Challenges, Remaining Work & Phase-III Roadmap

Challenges Faced

- CSV parsing and numeric conversion
- File path handling
- Usage-hour and tariff assumptions
- Understanding the dataset status field
- Collecting real test evidence

Solutions / Learning

- Validate CSV fields
- Check file paths
- Document assumptions
- Test outputs repeatedly

Phase-III Roadmap

1. Complete remaining modules
2. System integration
3. Testing / optimization
4. Final deployment / validation
5. Documentation

Brief Discussion: Main remaining tasks are reconciling the 100-record report dataset with the 20-appliance GUI view, completing calculation and CSV error tests, and capturing any additional evidence needed for final evaluation.

Challenges, Remaining Work & Phase-III Roadmap – Scope

Current Limitations

- Appliance data is predefined rather than measured live.
- Daily operating hours are assumed.
- Estimated cost depends on the configured tariff.
- The status field may be simulated; confirm its actual effect in the code.
- Rule-based insights, if present, are not machine-learning predictions.

Possible Future Work

- Editable tariff and budget values
- Historical usage comparison
- Stronger input validation
- Approved meter-data integration
- Appliance scheduling only after implementation

The current GUI screenshot set shows 20 appliances over 30 days, while the report describes a 100-record CSV. The final submitted build and dataset should be aligned. These future items are not claims about current functionality.

Expected Outcomes & References

Expected Final Outcomes

- GUI for appliance and energy views
- Estimated energy and cost summaries
- Budget and weekly insight views
- C++/CSV processing documentation
- Completed validation and final report

References

1. Project source code and CSV dataset
2. C++ documentation actually consulted
3. Official documentation actually used
4. Books or research papers actually consulted
5. GitHub repository: `https://github.com/ruhanianeja29/Electricity-consumption-analyzer`

Brief Discussion: The project combines C++ processing and a web GUI to summarize simulated household energy use. Completion is estimated at 70%; testing is partial.

Thank You

Questions & Suggestions